Domain and Range of Functions

Algebra Worksheet · Grade 8-10

Name: ______________________________ Date: _______________ Score: / 10

Learning Objectives

- Identify the domain and range of a relation given as a set of ordered pairs
- Determine the domain and range of linear and non-linear functions from equations
- Use graphical reasoning (vertical and horizontal sweep) to find domain and range

For each problem, find the domain and range of the given relation, equation, or function and express your answer using set notation or interval notation.

1. Find the domain and range of the relation given by the set of ordered pairs.

$$\{(3,4),\ (7,5),\ (8,3),\ (5,1)\}$$

Answer: ________________________________

2. Find the domain and range of the linear function.

$$y = 3x + 1$$

Answer: ________________________________

3. Find the domain and range of the linear function.

$$y = -2x + 5$$

Answer: ________________________________

4. Find the domain and range of the quadratic function.

$$y = x^2$$

Answer: ________________________________

5. Find the domain and range of the square root function.

$$y = \sqrt{x}$$

Answer: ________________________________

6. Find the domain and range of the rational function.

$$y = \frac{1}{x}$$

Answer: ________________________________

7. Find the domain and range of the relation.

$$\{(-2,7),\ (0,-1),\ (4,7),\ (6,3)\}$$

Answer: ________________________________

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8. Find the domain and range of the absolute value function.

$y = |x|$

Answer: ____________________________________

9. Find the domain and range of the constant function.

$y = 4$

Answer: ____________________________________

10. Find the domain and range of the shifted quadratic function.

$y = (x - 2)^2 + 3$

Answer: ____________________________________

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ANSWER KEY & SOLUTIONS

Encourage students to visualize sweeping a ruler horizontally for domain and vertically for range when working with graphs.

Solutions

1. Find the domain and range of the relation given by the set of ordered pairs.

$\{(3,4),\ (7,5),\ (8,3),\ (5,1)\}$

→ *List the first coordinate (x-value) from each ordered pair: 3, 7, 8, 5.*
→ *List the second coordinate (y-value) from each ordered pair: 4, 5, 3, 1.*
→ *Write the domain set in increasing order and the range set in increasing order.*

Answer: $\text{Domain} = \{3,5,7,8\},\ \text{Range} = \{1,3,4,5\}$

2. Find the domain and range of the linear function.

$y = 3x + 1$

→ *Sweep an imaginary vertical ruler across the x-axis; the line is touched at every x-value.*
→ *Sweep an imaginary horizontal ruler up the y-axis; the line is touched at every y-value.*
→ *Conclude that both the domain and range are all real numbers.*

Answer: $\text{Domain} = \mathbb{R},\ \text{Range} = \mathbb{R}$

3. Find the domain and range of the linear function.

$y = -2x + 5$

→ *Recognize that this is a non-vertical line with a defined slope.*
→ *A non-vertical line extends infinitely in both x and y directions.*
→ *State both the domain and range as the set of all real numbers.*

Answer: $\text{Domain} = \mathbb{R},\ \text{Range} = \mathbb{R}$

4. Find the domain and range of the quadratic function.

$y = x^2$

→ *Any real number can be squared, so the domain is all real numbers.*
→ *The smallest output of x squared is 0, achieved when x equals 0.*
→ *Outputs grow without bound upward, so the range is from 0 to infinity, including 0.*

Answer: $\text{Domain} = \mathbb{R},\ \text{Range} = [0,\infty)$

5. Find the domain and range of the square root function.

$y = \sqrt{x}$

→ *The expression under the square root must be non-negative, so x must be at least 0.*
→ *The principal square root returns only non-negative outputs.*
→ *Both domain and range start at 0 and extend to positive infinity.*

Answer: $\text{Domain} = [0,\infty),\ \text{Range} = [0,\infty)$

6. Find the domain and range of the rational function.

$y = \frac{1}{x}$

→ *Division by zero is undefined, so x cannot equal 0.*
→ *The graph never crosses the x-axis, so y cannot equal 0 either.*
→ *Exclude 0 from both the domain and the range.*

Answer: $\text{Domain} = \{x \in \mathbb{R} \mid x \neq 0\},\ \text{Range} = \{y \in \mathbb{R} \mid y \neq 0\}$

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7. Find the domain and range of the relation.

$\{(-2, 7),\ (0, -1),\ (4, 7),\ (6, 3)\}$

→ *Collect all first coordinates: -2, 0, 4, 6.*
→ *Collect all second coordinates and remove duplicates: -1, 3, 7.*
→ *Write each set in increasing order.*

Answer: $\text{Domain} = \{-2, 0, 4, 6\},\ \text{Range} = \{-1, 3, 7\}$

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8. Find the domain and range of the absolute value function.

$y = |x|$

→ *The absolute value is defined for every real number.*
→ *Absolute value outputs are never negative; the minimum value is 0.*
→ *Therefore the domain is all reals and the range is 0 to infinity.*

Answer: $\text{Domain} = \mathbb{R},\ \text{Range} = [0, \infty)$

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9. Find the domain and range of the constant function.

$y = 4$

→ *The horizontal line is defined for every value of x.*
→ *Every input produces the same output of 4.*
→ *So the domain is all real numbers and the range is the single value 4.*

Answer: $\text{Domain} = \mathbb{R},\ \text{Range} = \{4\}$

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10. Find the domain and range of the shifted quadratic function.

$y = (x-2)^2 + 3$

→ *The squared expression accepts any real x, so the domain is all real numbers.*
→ *The minimum value of the squared term is 0, occurring when x equals 2.*
→ *Adding 3 raises the minimum output to 3, so the range starts at 3 and goes to infinity.*

Answer: $\text{Domain} = \mathbb{R},\ \text{Range} = [3, \infty)$